

Improving the Accuracy of DNA Characterisation: A Comparative Study of DNA-BERT and the Nearest Neighbour Model

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Abstract

Toehold-mediated strand displacement (TMSD) is useful for DNA circuitry, but predicting rate constants remains difficult because strand-displacement kinetics depend on interacting sequence, structural and environmental factors. Physical models connect rate behaviour to molecular mechanisms such as base pairing, nearest-neighbour interactions, toehold length, temperature and ionic conditions, while machine learning approaches learn predictive patterns from experimental data. This study builds on Akay’s use of DNA-BERT for strand-displacement rate prediction by testing whether frozen contextual embeddings become more useful when combined with explicit nearest-neighbour dinucleotide features and nucleotide availability values. The Akay dataset was processed into reusable embedding, nearest-neighbour and availability artefacts, then evaluated using a reproducible Python pipeline that stored model checkpoints, predictions, figures, and verification outputs. The selected hybrid regression model attained the strongest held-out performance, with test $R^2 = 0.702$, compared with $R^2 = 0.687$ for the optimised DNABERT-2 implementation. Ablation experiments showed that nearest-neighbour and availability features improved performance over embeddings alone, while robustness testing and grouped permutation analysis indicated that all three feature families contributed predictive information. External candidate-ranking comparison showed partial agreement with Akay’s reference rankings rather than exact reproduction. Overall, the results suggest that broader sequence representations contain useful predictive information for TMSD kinetic prediction. However, the study does not isolate the physical contribution of individual sequence regions, and predicted candidate sequences would still require experimental testing.

1 Introduction

1.1 Nucleotide Polymers

DNA strands store genetic information as polymers of nucleotides connected via a sugar-phosphate backbone.[2]

1.1.1 Sugar-Phosphate Backbone

The backbone of a strand is composed of phosphodiester bonds linking pentose sugars at the 3' and 5' positions. These bonds provide structural stability, while the negatively charged phosphate groups contribute to electrostatic repulsion along the strand.[2]

1.1.2 Nucleotides

Nucleotides are the fundamental building blocks of DNA and consist of a nitrogenous base, a pentose sugar, and a phosphate group. The nitrogenous bases can be divided into two structural classes: purines (adenine and guanine), which contain a fused double-ring structure, and pyrimidines (cytosine and thymine), which contain a single-ring structure. Nucleotides polymerise through phosphodiester bonds to form DNA strands.[2]

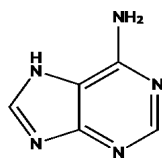


Figure 2: Adenine

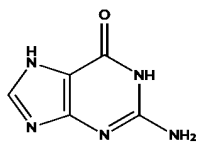


Figure 3: Guanine

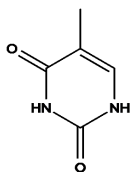


Figure 4: Thymine

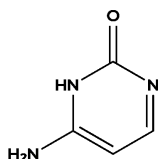


Figure 5: Cytosine

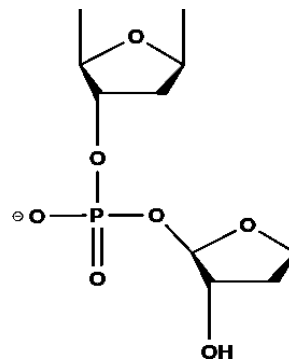


Figure 1: Sugar-Phosphate Backbone

1.1.3 Base Pairing

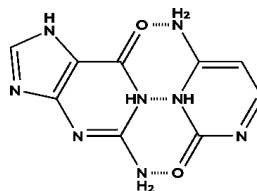


Figure 6: Guanine and Cytosine

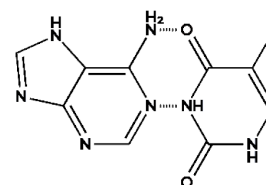


Figure 7: Adenine and Thymine

Complementary nucleotide bases on opposing DNA strands form base pairs through hydrogen bonding. Adenine and thymine are capable of forming two hydrogen bonds, whereas cytosine and guanine can form three hydrogen bonds. GC-rich duplexes are therefore often more thermally stable, although stacking interactions, temperature and ionic conditions also contribute to duplex stability.[2, 6]

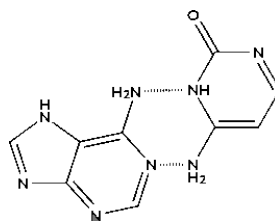


Figure 8: Mismatched Base Pair (AC)

Mismatched base pairs can cause reduced stability and altered kinetic behaviour via the partial exposure of a base's binding site to the environment.

1.2 Stacking Interactions

Strand stability and binding kinetics are influenced by sequence-dependent stacking interactions between neighbouring base pairs, which contribute to the local stability of a pair. This context dependence forms the basis of the nearest-neighbour model, which approximates the thermodynamic properties of base pairs using experimentally obtained parameters from local dinucleotide interactions.[6]

1.3 Molecular Thermodynamics

$$\underbrace{\Delta G}_{\text{Gibbs free energy}} = \underbrace{\Delta H}_{\text{enthalpy}} - \underbrace{T}_{\text{temperature}} \cdot \underbrace{\Delta S}_{\text{entropy}}$$

1.4 DNA Strand Kinetics

1.4.1 Stability and Kinetics

Environmental factors such as temperature and ionic strength can also affect association and dissociation rates. Association and dissociation occur reversibly, with their relative rates determining the equilibrium distribution of bound and unbound strands.[7]

1.4.2 Bond Dissociation (k_2)

Bond dissociation occurs when hydrogen bonds between base pairs break down. The rate of this reaction is influenced by ΔG , with negative ΔG values indicating a more stable duplex and therefore a slower dissociation rate.[7]

1.5 Toehold-mediated Strand Displacement (TMSD)

TMSD is a reaction pathway that uses toehold binding and branch migration to control strand exchange between DNA complexes.[5]

1.5.1 Toehold Mediation

A toehold is a short reactive DNA segment that provides the initial binding site for an invading strand during strand displacement. Toeholds are typically 6-10 nucleotides long, though other lengths are possible. Their length and sequence composition influence reaction kinetics and duplex stability, with AT-rich toeholds generally forming less stable duplexes and dissociating more readily.[8]

1.3.1 Gibbs Free Energy (ΔG)

Gibbs free-energy change indicates whether duplex formation is thermodynamically favourable under certain conditions.

1.3.2 Enthalpy (ΔH)

Enthalpy describes the energetic contribution of molecular bonds during binding.

1.3.3 Entropy (ΔS)

Entropy describes the number and distribution of accessible molecular configurations, which changes when strands bind to form a duplex.[6]

1.4.3 Bond Association (k_1)

Association occurs when complementary strands bind through base-pair formation. The Gibbs free-energy change, ΔG , describes the thermodynamic favourability and stability of the resulting duplex. A more negative ΔG favours the bound state, but does not necessarily indicate faster association, as the association rate also depends on kinetic barriers, sequence accessibility, temperature, and ionic conditions.[7]

1.5.2 Branch Migration

Branch migration replaces base pairs on an incumbent strand with the base pairs of an invading strand. Upon completion, the incumbent strand is released as a waste by-product. When initiated via toehold binding, branch migration serves as a basis for DNA circuitry as multiple displacement reactions can be arranged sequentially to form logical circuits.[9]

1.5.3 Circuit Timing

Nucleotide polymer binding kinetics are influenced by the environment; factors such as ionic strength and temperature may impede the functionality of TMSD circuits, which rely on precise timing.[7]

1.6 Objectives

1.6.1 Toehold Rate Constant Prediction

Improved prediction of toehold binding rate constants would support the development of robust TMSD implementations by supporting more reliable kinetic characterisation.[1]

1.6.2 Reproducible Framework

Because this study compared several feature sets and model variants, reproducibility was built into the workflow. Intermediate artefacts, model checkpoints, prediction outputs and verification records were stored so reported results could be regenerated with the same configuration.

1.6.3 Hybrid Feature Embedding

Akay’s results provided the starting point for this study: DNA-BERT embeddings appeared useful for predicting strand-displacement kinetics, but embeddings alone do not explicitly encode nearest-neighbour structure or nucleotide availability. This study, therefore, tested whether combining these feature types could improve TMSD rate prediction.[1]

1.7 Related Work

Existing TMSD modelling gives two useful perspectives on the same problem. Physical models explain why strand-displacement rates depend on base pairing, nearest-neighbour interactions, toehold length, branch migration, temperature and ionic conditions.[6, 7, 8] These models are important because they connect rate behaviour to molecular mechanisms, but exact prediction is still difficult because the factors do not act independently. Machine learning approaches take a different route by learning predictive patterns from experimental data. Akay showed that DNA-BERT embeddings can predict rate constants, suggesting that learned DNA sequence representations contain information relevant to TMSD kinetics.[1, 4] This study builds on that result by asking whether frozen DNA-BERT embeddings become more useful when combined with explicit nearest-neighbour features and nucleotide availability values.

2 Materials

2.1 Akay Dataset

The experimental data came from the published tables in the Akay paper.[1] These tables provided measured strand-displacement sequences, association rate constants and nucleotide availability values for supervised learning. The p53-related candidate table, which contained candidate sequences rather than measured training labels, was kept separate from training and used after model selection to compare the selected model’s ranking against Akay’s reference predictions.

2.3 Tokeniser and Checkpoint

To avoid training a transformer from scratch, this study used the DNA-BERT tokeniser and pre-trained Hugging Face checkpoint locally. The checkpoint weights were used only to generate fixed embeddings for the Akay sequences, while the downstream regression models were trained on the TMSD data.[4]

2.2 DNA-BERT

DNA-BERT was used as the main source of contextual sequence embeddings. Rather than representing a strand only as bases or counted motifs, DNA-BERT converts tokenised sequences into embedding vectors whose values depend on the surrounding context. In this study, those vectors were fixed input features for the regression models.[3]

2.4 DNABERT-2

DNABERT-2 was included as a comparison model because it represents DNA differently from the original 6-mer DNA-BERT model. DNABERT-2 has no direct 6-mer equivalent and instead uses a learned subword vocabulary. It replaces fixed k-mer tokenisation with byte-pair encoding, so tokens are learned from sequence statistics rather than fixed biological windows.[10]

3 Methodology

3.1 Pipeline Framework

Because multiple feature configurations and model variants were evaluated, the study required a pipeline that preserved consistent data interpretation, feature ordering, model storage and reproducibility checks. The workflow was built around Python abstractions with stable inputs and outputs for feature generation, training, prediction and verification. ExperimentConfig linked the artefact paths, dataset schema, reproducibility procedures and model settings used throughout the study.

3.1.1 Data Storage

Generated embeddings were frozen as NPZ artefacts, a NumPy format for storing named arrays, containing sequence identifiers, embedding vectors and metadata. Frozen embeddings meant that DNA-BERT weights were not updated during regression-head training. This separated transformer inference from model training, allowing embeddings to be reused across experiments. Prediction outputs were also stored as artefacts containing candidate identifiers, predicted rate constants, model metadata, and feature-configuration details.

3.3 Feature Generation

3.3.1 DNA-BERT Embeddings

DNA sequences were 30 nucleotides long and were tokenised into 25 overlapping 6-mer subsequences, corresponding to the representation used by the pre-trained DNA-BERT model. Special tokens required by the transformer architecture were added during tokenisation, and sequences were padded as needed to ensure a consistent input format for the encoder. The embedding-generation stage converted validated sequence records into reusable embedding artefacts. Mean pooling averaged non-padding token hidden states into one 768-dimensional embedding per sequence. Padding tokens were excluded to avoid sequence-length artefacts. Although the tokeniser can represent the ambiguity symbol N, no reported experimental sequences contained ambiguous nucleotides, and all nearest-neighbour features were generated from canonical bases only.

3.2 Dataset Pre-processing

The Akay dataset was published as PDF tables, so pdfplumber was used to extract the data into CSV format. Extracted tables were validated for malformed records, missing rows and missing columns. DNA sequences were uppercased, stripped of whitespace and checked against the supported bases A, C, G and T.[1] This validation step was handled via a dataset schema rather than embedded directly in the training code. The schema records which columns contain identifiers, sequences and availability values, while the DNA sequence validator normalises valid sequences and rejects unsupported nucleotide symbols. This made the interpretation of the Akay tables explicit throughout the study.

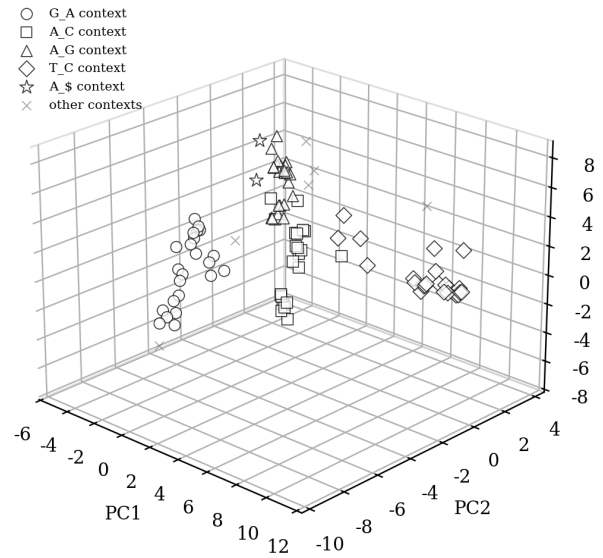


Figure 9: PCA projection of DNA-BERT hidden representations for the same 6-mer observed in different sequence contexts.

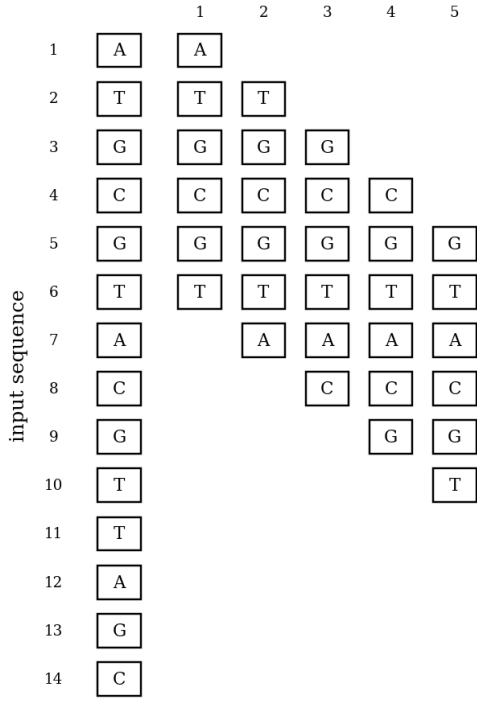


Figure 10: Overlapping 6-mer tokenisation window used to convert DNA sequences into DNA-BERT input tokens.

3.3.2 Nearest-Neighbour Vector Calculations

Nearest-neighbour features were derived from adjacent dinucleotide positions. A 30-base sequence contains 29 adjacent dinucleotides, each represented as one of 16 canonical states. Each position was encoded using one-hot encoding, producing a 16-element vector for every dinucleotide position. The positional component preserves local sequence context by recording the identity of each adjacent dinucleotide pair at a specific location, while the frequency component captures broader sequence composition trends. Together, these representations provide both local and global descriptions of dinucleotide structure. One-hot encoding was selected as it represents dinucleotide identities as categorical states without imposing an artificial numerical ordering between categories. The positional representation produced a 464-dimensional feature vector. A separate frequency representation was generated by calculating the normalised occurrence of each of the 16 dinucleotide states across the sequence, producing a further 16-dimensional feature vector. The selected configuration used the nearest-neighbour feature mode “both”, which

combines positional and frequency representations into a single 480-dimensional nearest-neighbour feature vector.

3.3.3 Nucleotide Availability Vectors

Nucleotide availability values were extracted directly from the processed Akay dataset, converted into numeric feature arrays, and aligned with their corresponding DNA sequences using sequence identifiers. These values were treated as position-level accessibility features supplied by the Akay dataset. As all reported sequences were 30 nucleotides long, each sequence produced a 30-dimensional availability feature vector. The processing pipeline accepts continuous numeric values and does not restrict availability features to binary states. Unlike the nearest-neighbour representation, availability values were not derived from sequence composition; they were imported from the dataset as an additional experimentally informed feature source.

3.4 Alignment

Because the feature families used different numerical scales, DNA-BERT embeddings, nearest-neighbour feature vectors and availability vectors were standardised before concatenation. The target variable k_1 was transformed using $\log_{10}$. This transformation reduces the influence of extreme values and yields a distribution better suited to optimisation by gradient-based learning algorithms. Standardisation parameters were calculated from the training data and then reused on the validation and test sets to ensure a consistent feature representation and prevent information leakage from unseen samples.

3.5 Feature Validation and Assembly

Validation checks were performed to ensure that embedding records, nearest-neighbour feature vectors and availability vectors corresponded to the same DNA sequence. Records with missing identifiers or mismatched sequence assignments were rejected to prevent incorrect feature associations during training.

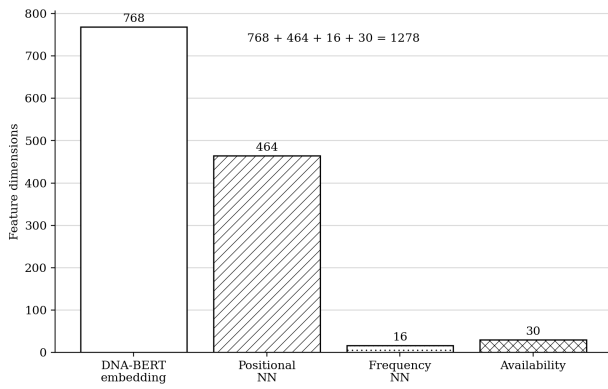


Figure 11: Total dimensions for each feature set used in model training.

Feature vectors were concatenated using NumPy to produce a unified feature representation for model training. The combined representation consisted of 768 DNA-BERT embedding dimensions, 480 nearest-neighbour dimensions, and 30 nucleotide availability dimensions, resulting in a total feature dimensionality of 1,278. The selected feature combination was stored as a FeatureRecipe, which records the feature families, the ordered feature names, the nearest-neighbour mode, and the standardisation parameters required to reconstruct the same input matrix. Feature ordering was preserved throughout training and inference to ensure that each model received a consistent representation of the input sequence.

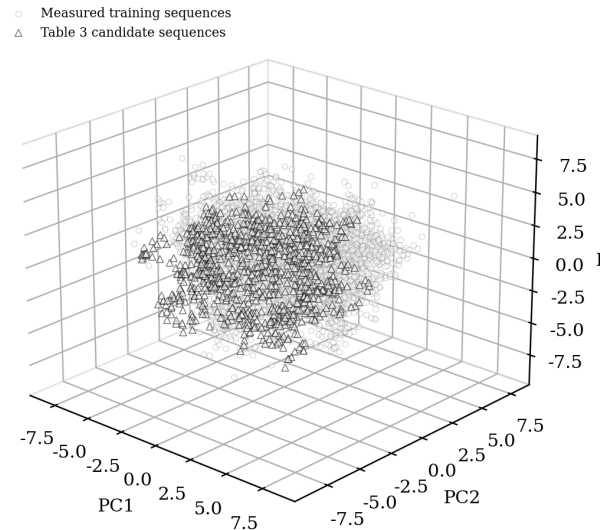


Figure 12: PCA projection of DNA-BERT embeddings for measured Akay Table 1 records and candidate Akay Table 3 records, showing that both datasets occupy a shared frozen-embedding space.

3.6 Dataset Split

The training set was used to optimise model parameters, the validation set was used for model selection and checkpoint evaluation, and the test set was reserved for final performance evaluation. Separating these datasets reduced the risk of overfitting by ensuring that parameter optimisation, model selection, and final evaluation were performed on distinct subsets of the data.

3.7 Model Training

The prediction model was a feed-forward regression head that accepted the prepared feature vector and output a transformed $\log_{10}(k_1)$ prediction. This architecture was selected for its simplicity, nonlinear modelling capacity and relatively low parameter count for the dataset size.

3.7.1 Regression Head

The regression head included linear layers, GELU activation functions, layer normalisation, and dropout regularisation. Linear layers perform the primary feature transformation, while normalisation stabilises feature distributions between layers.

3.7.2 Validation-based Model Selection

To prevent overfitting, the checkpoint with the lowest validation loss was selected for final evaluation rather than the final training epoch. This helps select a model that generalises better to unseen data rather than simply performing well on the training set. The selected checkpoint was stored with a ModelArtifact containing the model architecture parameters, FeatureRecipe and target transformation metadata required to reconstruct the same inference path.

3.8 Optimisations

Several optimisation methods were used, including AdamW, GELU activations, dropout regularisation and layer normalisation.

3.8.1 AdamW

The AdamW optimiser was used with mean squared error (MSE) for adaptive learning-rate behaviour and stable convergence during neural network training. This made it suitable for high-dimensional contextual embeddings and combined engineered feature vectors.

3.8.3 Dropout Regularisation

Dropout regularisation probabilistically disables subsets of neurons during training, reducing excessive memorisation of the training data and improving generalisation performance on unseen records.

3.8.2 GELU

GELU activation functions were used because they provide smoother nonlinear transformations than ReLU-based alternatives, allowing feature representations to be transformed without abrupt activation boundaries.

3.8.4 Layer Normalisation

Layer normalisation was incorporated to stabilise feature distributions between regression-head layers and improve optimisation consistency throughout training.

3.9 Feature-ablated Models

Feature-ablated models were trained to evaluate how each representation contributed to k_1 prediction. In all DNA-BERT 6-mer configurations, embeddings were generated from the pre-trained Hugging Face checkpoint and kept frozen, so only the downstream regression head was trained on the Akay k_1 labels. Figure 11 summarises the feature dimensions used by these configurations, while Figure 13 compares the primary 6-mer ablation sequence.

3.9.1 Embeddings only

The embeddings-only baseline used fixed 768-dimensional DNA-BERT contextual embeddings as the sole inputs to the regression head. No nearest-neighbour or nucleotide availability features were included, making this the reference point for evaluating later feature additions.

3.9.3 Embeddings + NN + Availability

The embeddings + nearest-neighbour + availability configuration added 30 nucleotide availability values from the processed Akay dataset, aligned to each sequence by identifier. The resulting 1,278-dimensional feature vector combined 768 DNA-BERT dimensions, 480 nearest-neighbour dimensions and 30 availability dimensions.

3.9.5 DNABERT-2 + NN + Availability

The unoptimised DNABERT-2 configuration used DNABERT-2 embeddings with the same nearest-neighbour and availability features as the selected hybrid model, achieving $R^2 = 0.647$. This provided a direct comparison between the original 6-mer DNA-BERT representation and DNABERT-2’s byte-pair encoded representation before additional DNABERT-2-specific tuning.[10]

3.9.2 Embeddings + NN

The embeddings + nearest-neighbour configuration concatenated the 768-dimensional DNA-BERT embedding vector with positional and frequency-based dinucleotide features, evaluating whether explicit local sequence structure improved prediction beyond the frozen contextual embedding alone.

3.9.4 Embeddings + NN + Availability (Optimised)

The optimised hybrid configuration used the same 1,278-dimensional feature vector with tuned regression-head settings. The selected architecture used fully connected layers, layer normalisation, GELU activations and dropout. Model selection retained the best validation checkpoint, which was then evaluated once on the held-out test set.

3.9.6 DNABERT-2 + NN + Availability (Optimised)

The optimised DNABERT-2 configuration used the same features but selected the best-performing DNABERT-2 regression-head configuration from a separate optimisation sweep. The best-performing optimised variant used a compact regression head with SiLU, a smooth nonlinear activation function, and achieved a test R^2 of 0.687.[10]

3.10 Result Storage and Scaling

Prediction outputs were stored as structured artefacts containing inference results, source paths, dataset schema and trace metadata. Results were saved in both transformed log space and raw k_1 scale. Raw-scale outputs express prediction errors in the original target units and support comparison with external predictions.

4 Workflow

Python dataclasses were used to keep workflow inputs and outputs explicit. `ExperimentConfig` recorded the dataset paths, schema, feature settings, model settings and reproducibility options for a run. The dataset schema defined how the extracted tables were interpreted, including the identifier, sequence, and availability columns. `FeatureRecipe` preserved the ordered feature families, the nearest-neighbour mode, and the standardisation parameters used to build the model input matrix. `ModelArtifact` linked the selected checkpoint to its feature recipe and target transformation metadata, while `ArtifactPredictionResult` stored generated predictions, candidate rankings and trace metadata.

The workflow began by extracting the Akay and p53-related candidate data from PDF tables, converting them to CSV, normalising sequences and aligning records using stable identifiers. Embedding generation and feature engineering then formed complementary branches: one branch tokenised sequences into overlapping 6-mers and generated DNA-BERT embeddings, while the other produced nearest-neighbour and nucleotide-availability vectors. `FeatureRecipe` recorded the resulting feature ordering and scaling so that the same representation could be reused during training, evaluation and candidate prediction. Model training optimised the regression head using transformed target values, monitored validation performance and retained the best checkpoint as a `ModelArtifact`. Output processing began with evaluation on the held-out test partition using MAE, RMSE and R^2 on transformed and raw scales. For candidate prediction, `PredictionService` received the selected `ModelArtifact` and dataset schema, reconstructed features, applied the saved scaling values, ran the regression head, inverted the target transformation, and ranked candidates by their predicted k_1 values.

The resulting `ArtifactPredictionResult` recorded the predictions, rankings and source metadata. External comparison evaluated agreement with Akay using Pearson correlation, Spearman correlation, top-k overlap, and within-2x agreement (the fraction of predictions falling within a two-fold difference), while reproducibility verification compared regenerated metrics, rankings, external comparisons, and rendered graphs against canonical artefacts.

4.1 Analytical Methodology

Model performance was evaluated using predicted k_1 values from the held-out test set, providing an estimate of generalisation to unseen data. Candidate sequences were ranked in descending order of predicted k_1 values. Regression performance was evaluated using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R^2 .

4.1.1 RMSE

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

RMSE measures the average magnitude of errors between predicted and actual values, with a greater penalty for larger errors; a higher RMSE indicates worse performance.

4.1.2 MAE

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

MAE measures the average absolute difference between predicted and actual values, providing a straightforward interpretation of prediction accuracy; a lower MAE indicates better accuracy.

4.1.3 R^2

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

R^2 measures the proportion of variance in the target variable that is predictable from the model; a higher R^2 indicates a stronger predictive fit.

5 Results

5.1 Evaluation

Evaluation was performed in two stages: internal test-set regression performance and external comparison against Akay candidate predictions. Internal performance was measured using MAE, RMSE, and R^2 , while external agreement was assessed using Pearson correlation, Spearman correlation, top-k overlap, and within-2x agreement.

5.2 Model Performance

Model configuration	R^2	RMSE	MAE
Embeddings only	0.240	906,381.063	601,778.375
Embeddings + NN features	0.335	848,327.750	588,749.063
Embeddings + NN + availability	0.626	635,703.000	406,633.500
Selected hybrid model	0.702	567,590.625	369,376.906
Unoptimised DNABERT-2	0.647	617,602.875	395,669.844
Optimised DNABERT-2	0.687	581,798.125	381,525.688

Table 1: Raw-scale performance of evaluated model configurations on the held-out test set. Lower RMSE and MAE values indicate smaller prediction errors, while higher R^2 values indicate a stronger predictive fit.

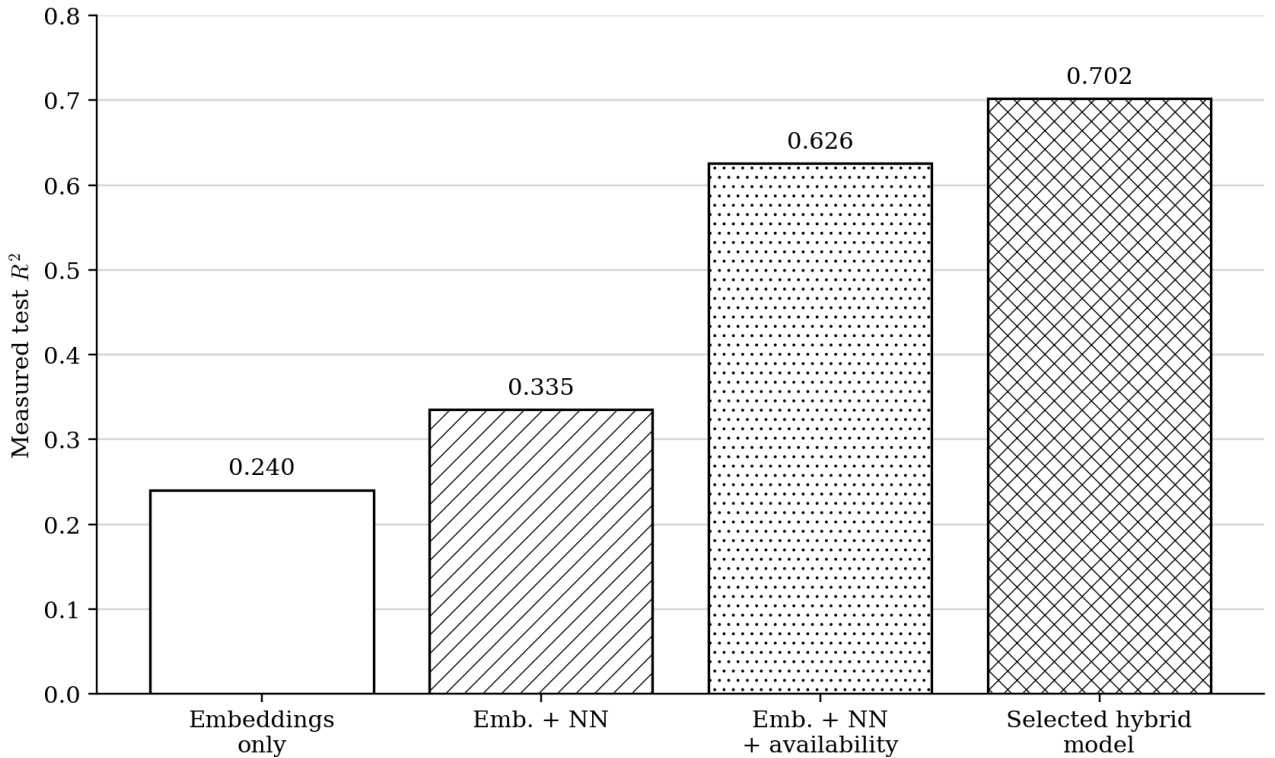


Figure 13: Feature ablation results showing the embeddings-only reproduction baseline and subsequent feature additions. Frozen DNA-BERT embeddings alone achieved a lower predictive fit, whereas nearest-neighbour and availability features improved the held-out test-set R^2 .

The ablation results in Figure 13 show that each feature addition improved the held-out predictive fit. The embeddings-only baseline achieved a test R^2 of 0.240, indicating that frozen contextual embeddings contained useful sequence information but were insufficient to match the hybrid configurations. Adding nearest-neighbour features improved test R^2 from 0.240 to 0.335, while adding nucleotide availability features increased performance to $R^2 = 0.626$. The selected hybrid model achieved the

strongest result, reaching test $R^2 = 0.702$, and also outperformed the optimised DNABERT-2 configuration, which achieved test $R^2 = 0.687$. This indicates that the fixed 6-mer DNA-BERT representation remained competitive when combined with engineered sequence and availability features.

Although aggregate predictive performance improved, the hybrid representation did not produce close agreement with Akay among the highest-ranked candidate sequences. Improvements were instead observed across the overall prediction task, suggesting that the hybrid representation improved regression performance more clearly than exact top-rank agreement with Akay.

Figure 14 shows that predicted values broadly follow the measured $\log_{10}(k_1)$ values, although dispersion remains visible in lower-rate regions of the test set.

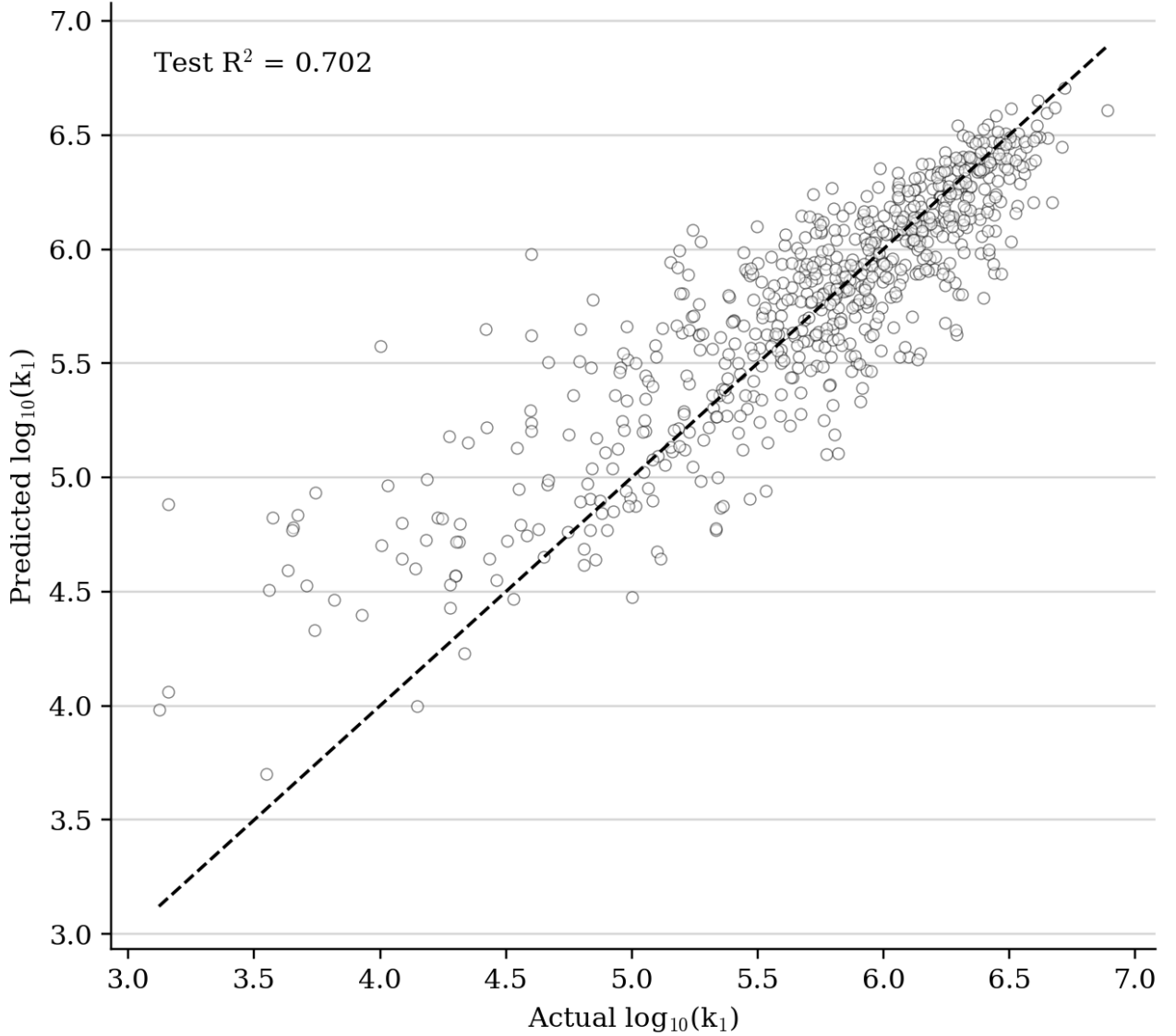


Figure 14: Predicted versus actual $\log_{10}(k_1)$ values for the selected 6-mer model on the held-out test set.

5.3 Error Analysis

Residual analysis was used to examine where the selected model made larger prediction errors, rather than only reporting aggregate performance. On the held-out test set, the median absolute residual was 0.164 in $\log_{10}(k_1)$ space, while the 90th percentile absolute residual was 0.533. The lowest measured-rate third had a larger mean absolute residual (0.345) than the middle (0.192) and highest (0.175) thirds, indicating that prediction errors were not uniformly distributed across the target range.

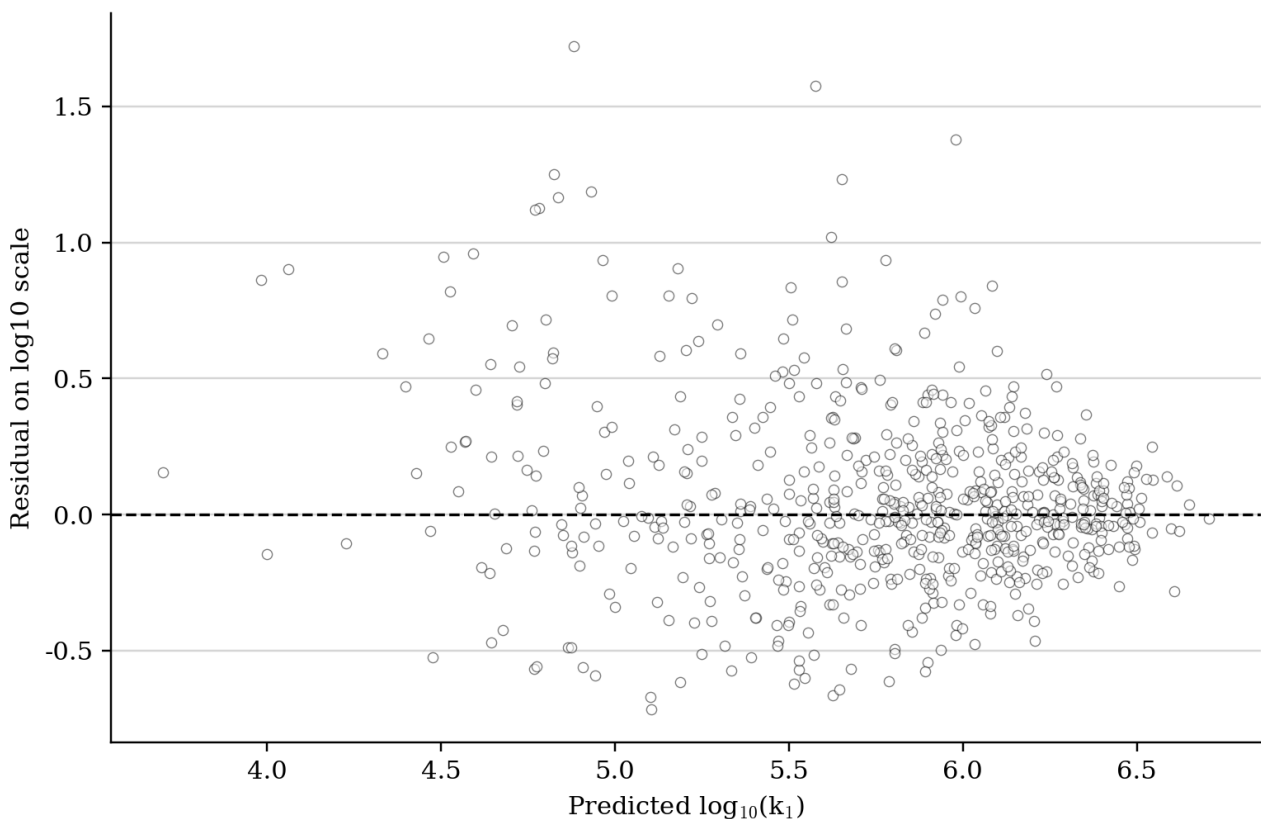


Figure 15: Residuals for selected-model $\log_{10}(k_1)$ predictions on the held-out test set, where residual = predicted - actual.

The residual pattern suggests that the selected model was less stable for lower-rate examples in the held-out test set. This does not identify the physical cause of individual errors, but it indicates that aggregate test performance should be interpreted alongside the distribution of prediction errors across kinetic regimes.

5.4 Robustness Regarding Random Seeds

To assess sensitivity to the train-validation-test partition, the selected hybrid configuration was repeated using three random seeds. Each run used the same feature configuration and training settings, varying only the data split and optimisation seed. The selected model remained consistent across these runs, with a mean test R^2 of 0.696 and a standard deviation of 0.010. Table 2 summarises the variation observed across the repeated selected-model runs.

Metric	Mean $\pm$ SD
R^2	0.696 ± 0.010
RMSE	$565,634 \pm 4,012$
MAE	$368,926 \pm 4,663$

Table 2: Robustness of the selected hybrid model across random seeds 7, 13 and 23.

5.5 Feature-Family Permutation Importance

Grouped permutation analysis was used to estimate the extent to which the trained, selected model depended on each feature family. For each feature family, the corresponding test-set columns were shuffled across samples while the remaining features were left unchanged. The model was then re-evaluated over 20 shuffles, and the mean decrease in raw-scale test R^2 was recorded.

Shuffling nucleotide availability features produced the largest reduction in predictive performance, followed by nearest-neighbour features and DNA-BERT embeddings. These results support the ablation findings by showing that the trained model used all three feature families, with the strongest test-set dependence on availability features. Because permutation disrupts the learned relationship between correlated feature groups and the target, these values should be interpreted as model-dependent measures rather than causal estimates of physical importance.

Feature family	Features	Mean R^2 drop
Availability	30	0.693
Nearest-neighbour	480	0.443
DNA-BERT	768	0.176

Table 3: Grouped permutation importance for the selected hybrid model on the held-out test set.

5.6 External Candidate Ranking

External ranking assessed whether the selected model prioritised candidate sequences similar to the Akay reference predictions. The comparison showed partial ranking agreement rather than exact reproduction of the Akay ordering, with agreement improving when broader top-ranked candidate sets were considered.

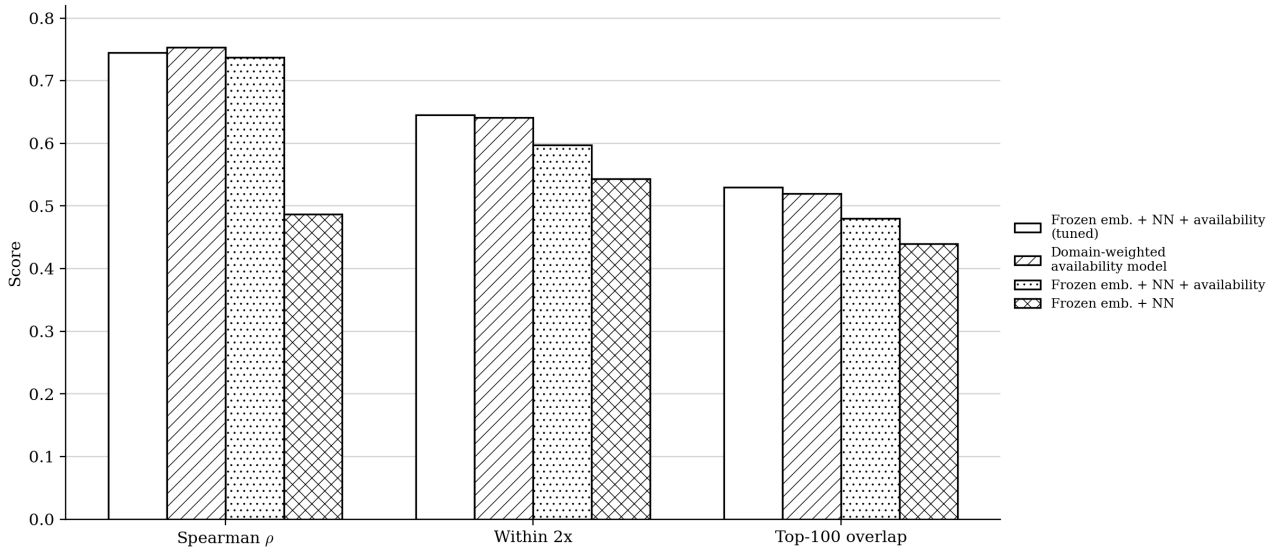


Figure 16: External ranking agreement metrics comparing selected model predictions with Akay reference predictions. The metrics evaluate candidate-prioritisation agreement rather than exact kinetic equivalence.

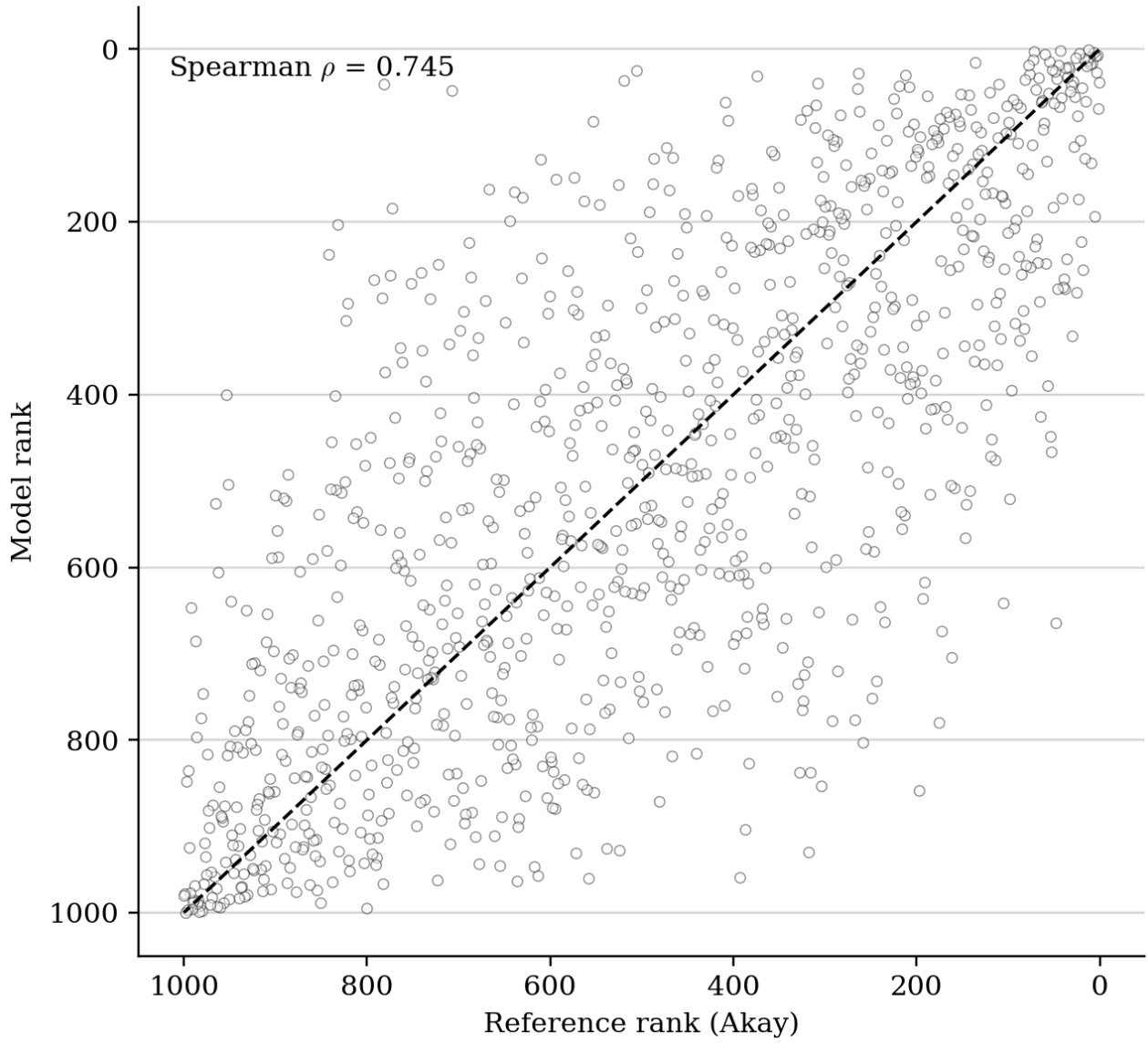


Figure 17: Rank agreement between selected model predictions and Akay reference rankings, showing broad candidate-prioritisation agreement rather than exact rank matching.

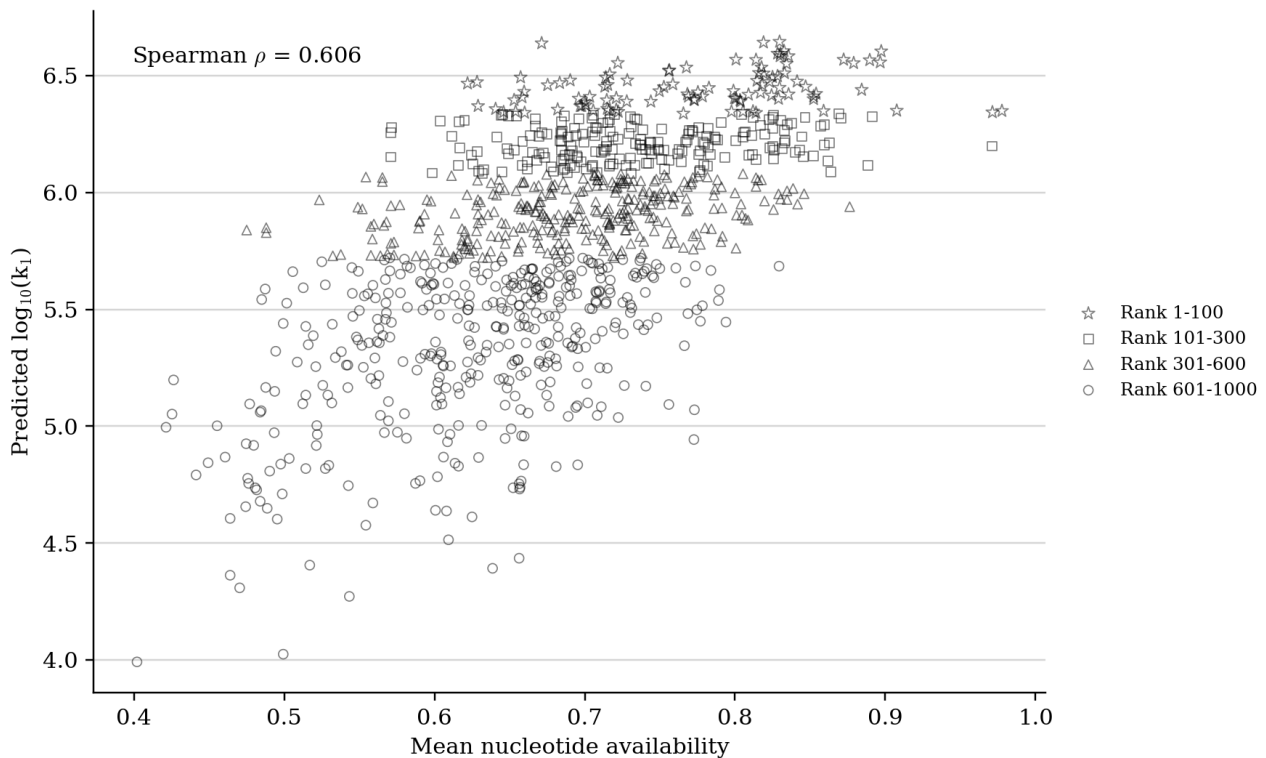


Figure 18: Relationship between mean nucleotide availability and predicted $\log_{10}(k_1)$.

The relationship between nucleotide availability and predicted rate was examined to assess whether the model's predictions were associated with the supplied availability features. Within the model outputs, greater mean availability was associated with higher predicted association rates. However, this relationship does not demonstrate a causal effect on experimentally measured kinetics.

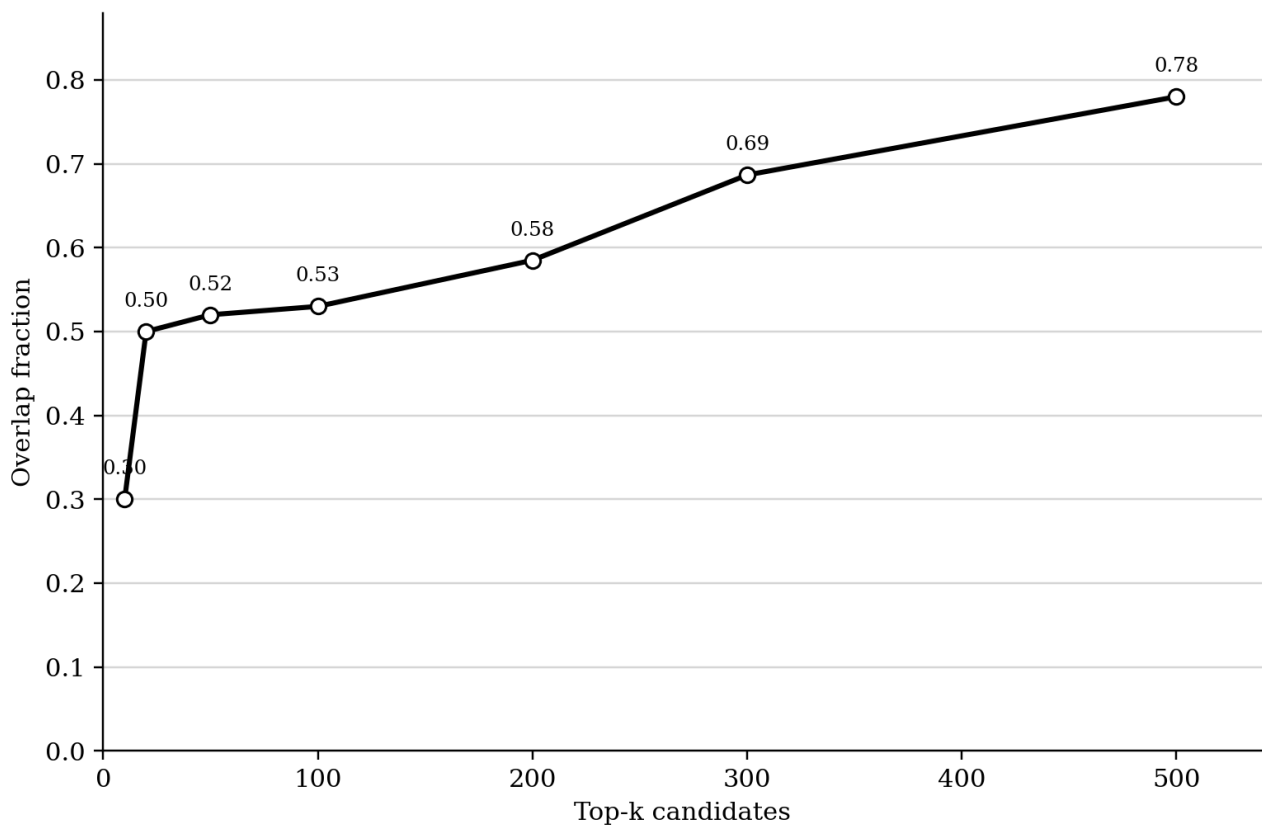


Figure 19: Top-k overlap between the selected 6-mer model and Akay reference predictions. The curve shows the proportion of shared candidates within the top-ranked prediction sets as k increases, measuring agreement in candidate prioritisation rather than exact predicted rate constants. Higher overlap values indicate stronger agreement between the two ranking approaches.

Candidate-ranking agreement provides a complementary, application-oriented evaluation alongside regression metrics. In practice, the model is intended to identify promising candidate sequences rather than precisely estimate k_1 for every individual strand. Candidates are sorted by predicted k_1 values, allowing the most promising sequences to be selected for further experimental testing.

6 Discussion

6.1 Local Nucleotide Interactions

Nearest-neighbour thermodynamics are based on the principle that the stability of a nucleotide pair is influenced by neighbouring bases rather than being determined in isolation. The improvement observed after adding nearest-neighbour features is consistent with this principle, suggesting that local dinucleotide context contributes useful predictive information for strand-displacement kinetics. DNA-BERT embeddings and nearest-neighbour features appear to provide complementary representations: the former captures broader contextual sequence patterns, while the latter explicitly encodes local dinucleotide structure. This result does not establish how individual interactions physically contribute to strand-displacement kinetics, as controlled ablation studies and experimental measurements would be required to determine the contribution of particular sequence regions or interaction types.

6.3 Limitations

The study was limited by its dependence on a single published dataset and did not include new wet-lab validation of the predicted candidate sequences. The nucleotide availability features were taken from the Akay dataset rather than generated independently, so the results evaluate their usefulness as supplied features rather than validating the availability-generation method itself. The ablation experiments show performance changes after adding feature families, but they do not isolate the physical contribution of individual sequence regions or determine causal mechanisms behind individual prediction errors. The selected model predicts association rate constants for the available strand-displacement records and should not be interpreted as a full circuit-level simulator. TMSD circuit behaviour also depends on reaction networks, concentration, temperature, ionic conditions and experimental implementation details. The results therefore support improved kinetic characterisation within the evaluated data regime, while leaving experimental testing and broader circuit-level generalisation as future work.

6.2 Implications for DNA Circuitry

For DNA circuitry, the practical implication is that rate prediction may need to account for more than just the designed toehold region. The selected model benefited from contextual representations spanning the complete input sequence, indicating that full-sequence representations can contain predictive information not captured by isolated nucleotide identities or toehold-focused descriptions. However, the study does not isolate individual sequence regions or prove that nucleotides outside the toehold physically determine kinetic behaviour. This distinction matters for circuit design. If sequence context changes predicted kinetics, then sequence design and circuit timing cannot be treated as fully independent design problems. Modifications intended to encode information or alter circuit behaviour may also affect reaction timing, introducing an additional constraint in the development of predictable molecular computing systems.

6.4 Future Work

6.4.1 Larger Experimental Datasets and Hybrid Models

Larger datasets may make it practical to fine-tune DNA-BERT models directly on strand-displacement data, enabling the embedding model to learn TMSD-specific sequence representations rather than relying solely on frozen genomic pretraining. They would also allow multi-scale sequence representations to be tested, helping determine whether predictive information is concentrated in local dinucleotide context or longer sequence motifs. Future work should also experimentally validate top-ranked candidate sequences and evaluate similarity-aware or repeated-splitting strategies to test whether the reported performance holds under stricter generalisation settings.

6.4.2 Non-canonical Nucleotide Polymers and Synthetic Backbones

Improved prediction methods may reduce uncertainty, but the molecular substrate may still limit how predictable DNA-based computational systems can become. Future work could investigate non-canonical nucleotide polymers, synthetic base pairs, and alternative backbone chemistries, as modified nucleotide interactions may reduce contextual sequence effects or alter the thermodynamic properties that underlie kinetic variability. Rather than solely improving the prediction of sequence-dependent behaviour, future research may also explore whether alternative molecular systems can reduce the degree of sequence dependence itself using modelling approaches similar to those developed in this study.

7 Conclusion

Although accuracy was constrained by dataset size and DNA kinetic complexity, the results showed that combining contextual embeddings, nearest-neighbour features and availability features improved prediction performance. The selected hybrid model achieved the strongest predictive performance among the evaluated configurations, with a test R^2 of 0.702 and dependable performance across repeated random seeds. Error analysis showed that prediction errors were larger in lower-rate regions, while grouped permutation analysis indicated that all three feature families contributed to model performance, with the strongest dependence on availability features. External ranking comparison showed meaningful but incomplete agreement with Akay candidate prioritisation, supporting the model as a reproducible candidate-screening tool rather than a replacement for experimental testing. The reproducibility checks confirmed that the reported metrics, rankings and graph artefacts could be regenerated from the stored pipeline configuration. However, while the ablation and permutation results show predictive dependence on the feature families, they do not isolate the physical contributions of individual sequence regions or establish the causal mechanisms underlying the observed improvements. Future advances in experimental data collection, validation of candidate sequences and molecular design may further improve the predictability of DNA-based systems and contribute to more robust DNA nanotechnology and molecular computing platforms.

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